



Fighting viruses on the nanoscale with AFM

Understanding structure, function and host interaction of viruses in WaveMode, an off-resonance imaging technique

During the last century humans have endured multiple severe viral outbreaks including the Spanish flu and other influenza epidemics, HIV, SARS-CoV, MERS, Ebola and the recent SARS-CoV2 pandemic. During this period, science has made strides in understanding many different facets of virus outbreaks including viral infectivity, pathogenicity, and virulence, transmission, and propensity to cause an epidemic/pandemic. Modern techniques that include cell culture, infectivity assays, polymerase chain reactions, fluorescence microscopy, electron microscopy, and immunoassays have helped us gain a better understanding about the virus life cycle, viral replication, and the virus-host interaction. We now have a better grasp on the structure and function of the viral particle, viral genome, RNA or DNA expression levels, virus-host cell interaction and immune response. These advances have resulted in the development of vaccines and drugs to treat several different viruses.

Keywords – Atomic Force Microscopy, AFM, HSV-1 capsids, WaveMode, off-resonance imaging, Nanosurf Drive AFM

Introduction

Atomic force microscopy (AFM) is one of the newer techniques available for virus research. AFM is a cantilever-based technique that utilizes a sharp tip to interrogate surfaces at resolutions well below the optical diffraction limit. Beyond imaging, AFM is also a powerful tool for nano-mechanical probing and nano-manipulation. AFM thus offers a large toolbox of applications to address structure, function, and host interaction of viruses.

One of the primary advantages of AFM compared to electron microscopy is that it can operate on samples immersed in liquid. This empowers experiments on

living cells or cell organelles at physiologically relevant conditions. AFM imaging has been used to study functional/infective virus particles, to investigate the dimension and morphology and packaging of viral genomic material. Beyond imaging, AFM has been used for manipulation of single viruses by force spectroscopy to study early events of virus–host interactions.

This article reports on high-resolution imaging and force spectroscopy on virus capsids attached to both glass substrates and isolated cell nuclei. In this study, virus particles were deposited on a single cell using a hollow cantilever and the host cell response upon virus infection was studied.

High-resolution imaging and force spectroscopy of Herpes simplex virus 1 capsids

The capability of AFM to observe samples in aqueous environments allows observing proteins, nucleic acids and assemblies thereof in a state that is as close to physiological as possible. Namely, keeping samples such as Herpes simplex virus 1 (HSV-1) capsids fully hydrated allows observing their unperturbed structure as well as investigating their nanomechanics.

WaveMode imaging, a photothermally driven off-resonance operation mode has been utilised to visualize HSV-1 capsids. Adsorption of the HSV-1 capsids to a modified glass surface resulted in weakly attached capsids as the surface area of the capsids interacting with the substrate is rather small due to their shape. As a result, imaging requires gentle conditions not to displace the capsids during the imaging process. WaveMode with its gentle imaging conditions that result in minimal vertical and lateral forces acting on the capsids during imaging allowed reproducible and fast visualization of HSV-1 capsids.

WaveMode imaging reveals fundamental structural features of HSV-1 capsids, such as their size or geometry. Figure 1A shows AFM images of two HSV-1 capsids revealing their icosahedral overall structure and

the arrangement of capsomeres within the different facets of the icosahedral structure.

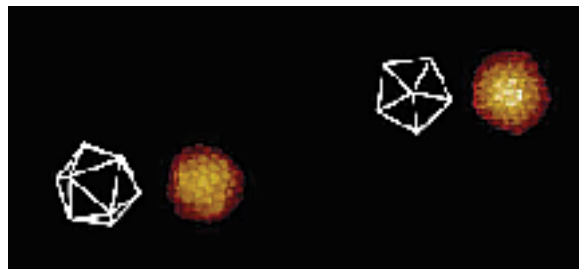


Figure 1A: AFM topography of two virus capsids adsorbed to a modified glass surface. The capsids adsorbed in different orientations. The white wireframes indicate the surfaces of the capsids facing upwards. Image width 940 nm, colour scale 55 nm.

The high resolution of the AFM also allows precisely measuring the diameter of such virus particles. For HSV-1, the measured height depends, because of the icosahedral structure, to some extent on the orientation in which the particles adsorbed to the surface. The examples in Figure 1B show a height of 115 nm.

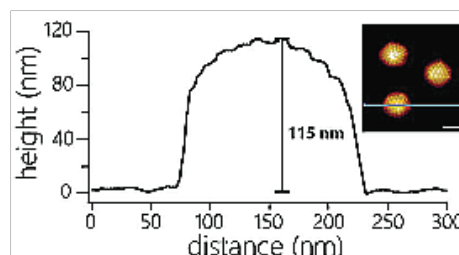


Figure 1B: Cross-sectional profile of one of the three capsids shown in the inset (scale bar 100 nm). The capsid shows an apparent height of 115 nm.

At slightly higher resolution, more details of the capsid structure can be resolved; for example, the center-to-center spacing of capsomeres can be determined (~17 nm; Figure 1C).



Figure 1C: 3D render of AFM topography of a virus particle at higher resolution than in figure 1A. The individual capsomeres can be observed and the center-to-center distance determined (17 nm).

After locating virus capsids with WaveMode imaging, individual capsids can be probed with the AFM probe to obtain the nanomechanical properties of these capsids. It is known, that the nanomechanical properties of virus capsids depend not only on the structure of the protein-based capsid but e.g. also on whether the capsids are filled with DNA or not¹.

To measure the nanomechanical properties of capsids, the cantilever is approached to the top of the capsid to apply an increasing force to the capsid. Such experiments result force-distance curves such as shown in Figure 2A. The force curves reveal a plethora of information, not only on the nanomechanics of the probed capsid. After contacting the capsid (Figure 2A, left dashed line) and upon application of force, the capsid deforms mainly elastically, as the force-distance curve shows a linear increase. The slope of the force distance curve (Figure 2A, orange line) reveals the spring constant of the capsid.

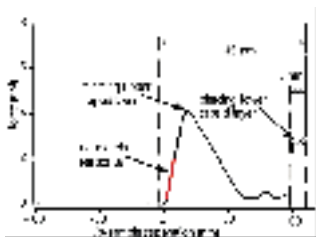


Figure 2A: Indentation curve on an HSV-1 capsid. The curve shows the elastic response (stiffness, orange line) of the capsid upon application of force, the penetration of the upper protein layer and the penetration of the lower capsid layer. The distinct features of the force curve also allow estimating the height of the capsid (115 nm) and the thickness of the protein layer (13 nm).



Figure 2B: 3D render of two capsids after indentation. The capsids were indented with the cantilever until both capsid layers were penetrated and re-imaged. The defect caused by indentation as well as capsomers not directly affected by the indentation can be clearly resolved. Image width: 705 nm.

As the force applied to the capsid increases, the capsid structure at some point fails and cannot withstand the

applied force anymore. The failure point is characterized by a sudden decrease of the force in the force-distance curve. The capsid structure failure occurs locally rather than causing complete capsid destruction: Re-imaging the capsid after penetration reveals a defect at the location where the cantilever interacted with the capsid (Figure 2B).

The force-distance curve also reveals information about the size of the capsid and the thickness of the capsid's protein shell. After piercing the first layer of proteins of the capsid, the tip is further approached towards the surface and at some point, meets the inside of the shell at the bottom of the capsid (Figure 2A, middle dashed line) and further movement loads the lower protein layer of the capsid from the inside. At sufficient force, also this protein layer is pierced and the tip pushes through and subsequently interacts with the underlying glass substrate (Figures 2A, right dashed line). The distance between the initial contact point with the capsid and the position at which the tip interacts with the glass surface reveals the capsid height at the point of indentation (~115 nm). The distance between contacting the lower protein layer and the interaction with the glass surface reveals the thickness of the capsid shell (~13 nm).

Imaging of HSV-1 capsids bound to isolated cell nuclei

In vivo, after entry into the cell HSV-1 capsids dock to nuclear pore complexes (NPCs) on the outer surface of the cell nucleus to release their DNA into the cell nucleus. Evilevitch and Tsimtsirakis established a reconstituted virus-nucleus system that allows investigating the nanomechanical changes taking place upon DNA injection and reorganization of chromatin inside the nucleus².

WaveMode revealed for the first time high-resolution topographic information of HSV-1 capsids on intact nuclei and not just spread, flat nuclear membranes. The AFM images reveal individual HSV-1 capsids bound to NPCs as well as NPCs not occupied by capsids. (Figure 3A). With WaveMode imaging, capsids bound to the soft

nuclei can be readily recognized by the individual capsomers that can be observed. (Figure 3B).

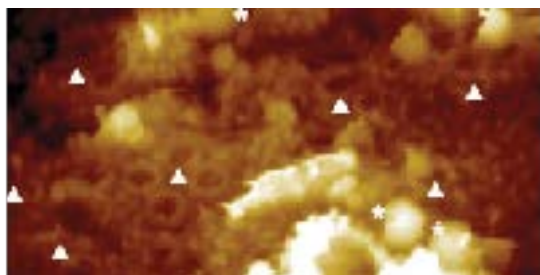


Figure 3A: Overview image of the surface of an intact isolated and fixed nucleus. The image reveals NPCs () and HSV-1 capsids (*) bound to NPCs. Image width 1650 nm.

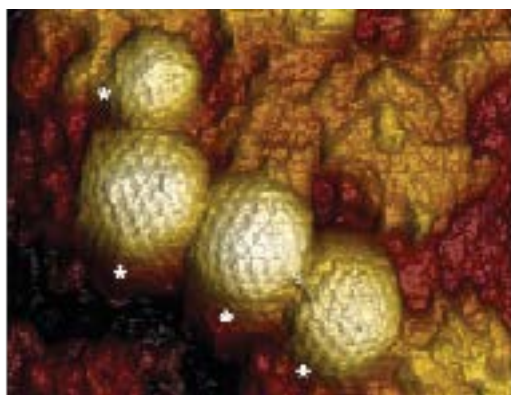


Figure 3B: 3D rendered AFM topography of multiple HSV-1 capsids bound to an intact nucleus. Image width 590 nm.

The reconstituted virus–nucleus system not only allows imaging NPC-bound HSV-1 capsids but also determining nanomechanical properties of nuclei under different conditions. Figure 4A shows a representative indentation curve on a nucleus incubated with DNA-containing HSV-1 capsids. Repeated indentation of multiple capsids reveals a Young's modulus of 214 ± 10 kPa (average $\pm$ SE) for nuclei incubated with C-capsids (Figure 4B).

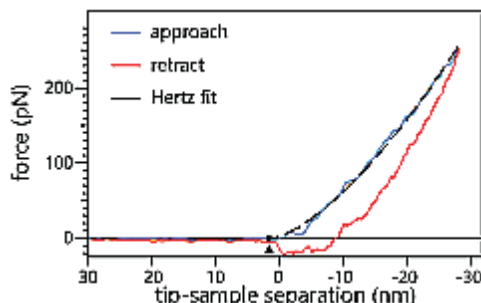


Figure 4A: Representative indentation curve on a non-fixed nucleus incubated with C-capsids. The dashed line represents the fit of the Hertz model to the data.

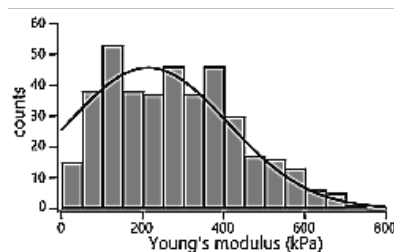


Figure 4B: Histogram of Young's moduli from repeated probing of multiple non-fixed intact nuclei incubated with C-capsids.

Conclusion

Virus research in the past has led to the development of treatments and vaccines. New viruses like the recent SARS-CoV2, which has led to the COVID-19 pandemic, stressing the importance of continued virus research to become a better and faster response to future outbreaks. AFM is a powerful technique to analyse structure and nanomechanics and manipulate biological samples under physiological conditions. It can be combined with advanced optical techniques, to allow correlation of optical characterization of the sample cells in parallel to the AFM experiments.

This application successfully explores the structure and nanomechanics of virus capsids using the WaveMode, an off-resonance imaging mode by Nanosurf Drive AFM.

References

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